Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 7

Tautologies, Contradictions, and Rules of Inference

Covers Chartrand & Zhang, Section 1.5

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first. Fill truth tables on this page.

Part I: How to Practice This Material

Three outcomes, not two

The classification

Build the final column. Then:

- all T ⇒ **tautology**
- all F ⇒ **contradiction**
- a mix ⇒ **neither**

"Neither" is the usual case. Do not force a statement into one of the two extremes because the problem seems to want it.

Three mistakes to expect

1. **Stopping after two rows.** A tautology claim needs every row.
2. **Mistaking a fallacy for a rule.** Affirming the consequent looks like modus ponens and is not a tautology.
3. **Forgetting that an equivalence is a tautology.** If $R \equiv S$ then $R \Leftrightarrow S$ is true in every row.

Part II: Practice Problems

Section A — Classify

1. For each, build a truth table and classify it as a tautology, a contradiction, or neither.

(a) $(P \Rightarrow Q) \vee (Q \Rightarrow P)$

(b) $(P \vee Q) \Rightarrow (P \wedge Q)$

(c) $\big(P \wedge (\sim Q)\big) \wedge \big((\sim P) \vee Q\big)$

2. Let S be a tautology and R a contradiction, built from the same components. Determine the truth value of each: (a) $S \vee R$; (b) $S \wedge R$; (c) $S \Rightarrow R$; (d) $R \Rightarrow S$; (e) $S \Leftrightarrow R$.

Section B — Rules and fallacies

3. For each argument, say whether the reasoning is valid. If it is, name the rule; if not, name the fallacy and give a situation where the premises hold but the conclusion fails.

(a) If the server is down, the page fails to load. The server is down. So the page fails to load.

(b) If the server is down, the page fails to load. The page failed to load. So the server is down.

(c) If the server is down, the page fails to load. The page loaded. So the server is not down.

(d) If the server is down, the page fails to load. The server is not down. So the page loaded.

4. Verify that modus tollens, $\big((P \Rightarrow Q) \wedge (\sim Q)\big) \Rightarrow (\sim P)$, is a tautology.

P	Q	$P \Rightarrow Q$	$\sim Q$	$(P \Rightarrow Q) \wedge (\sim Q)$	Whole statement
T	T				
T	F				
F	T				
F	F				

Section C — Building to order

5. Find a compound statement built from P and Q that is true in exactly 50% of the rows. Then find one built from P, Q, and R that is true in exactly 50% of its rows.

Section D — Writing

6. In five to eight complete sentences, explain why a rule of inference has to be a tautology, and why affirming the consequent is not one. Use the server example from Problem 3 concretely.

Section E — Challenge (optional)

7. Show that $\big((P \Rightarrow Q) \wedge (Q \Rightarrow R)\big) \Rightarrow (P \Rightarrow R)$ is a tautology. Then explain why this is *not* the same as claiming $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \equiv P \Rightarrow R$.